

Axial Compressor Theory and Design

Class 6

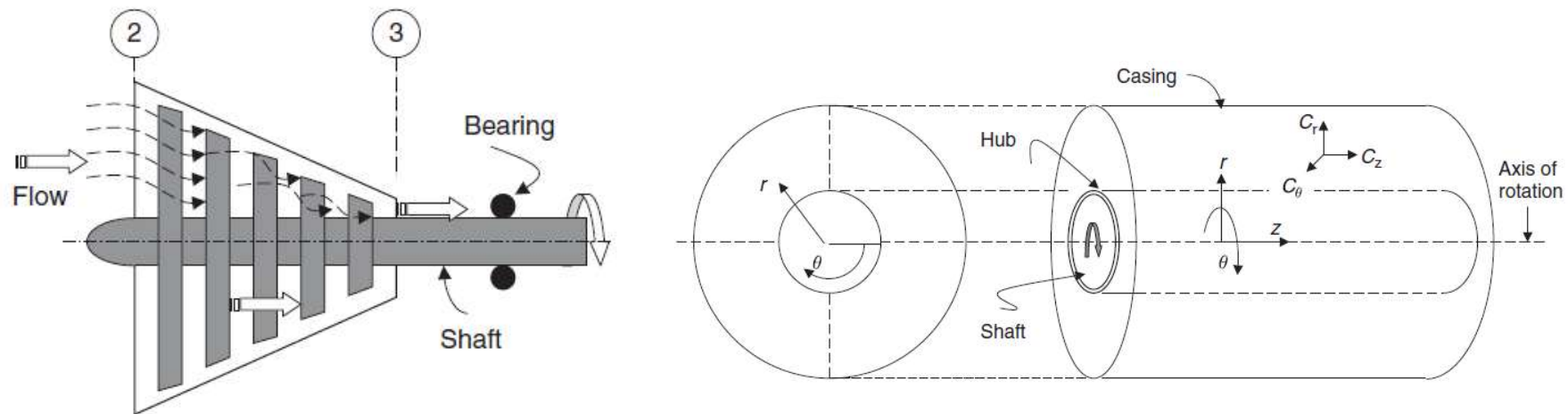
Introduction to the Design Process

- What is special about the design process?
 - Every designer has an image of what elements constitute the design process, and so our version is not likely to be exhaustive. Nevertheless, the following list contains critical elements with which few would disagree:
 - 1) The design process is both started by and constrained by an identified need.
 - 2) In the case of the design of systems, such as aircraft and engines, many legitimate solutions often exist, and none can be identified as unique or optimum. Systematic methods must be found to identify the most preferred or "best" solutions. The final selection always involves judgment and compromise.
 - 3) The process is inherently iterative, often requiring the return to an earlier step when prior assumptions are found to be invalid.
 - 4) Many technical specialties are interwoven. For example, gas turbine engine design involves at least thermodynamics, aerodynamics, heat transfer, combustion, structures, materials, manufacturing processes, instrumentation, and controls.
 - 5) Above all, the design of a complex system requires active participation and disciplined communication by everyone involved. Because each part of the system influences all of the others, the best solutions can be discovered (and major problems and conflicts avoided) only if the participants share their findings clearly and regularly.

Introduction

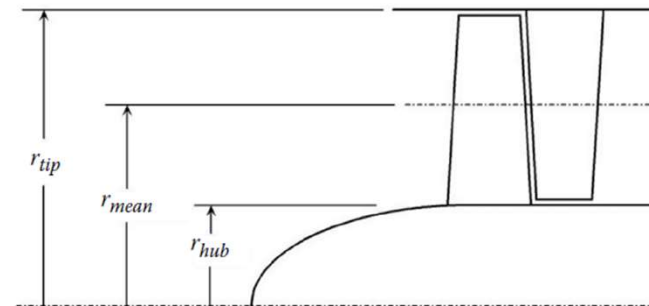
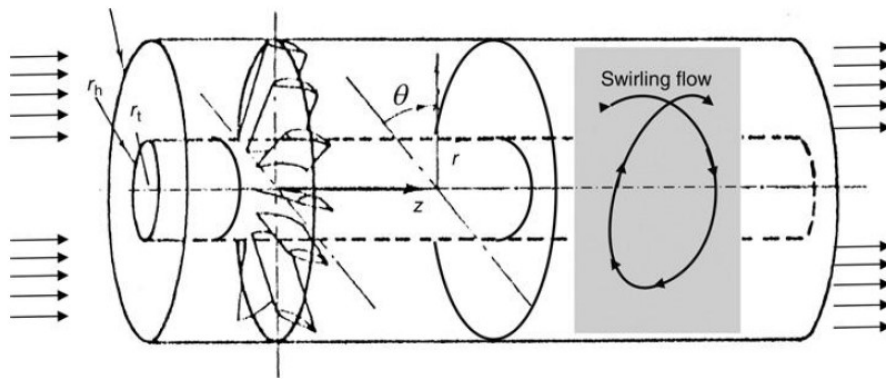
- Since day one of the gas turbine engine, the compressor was the key component for success and always required high development efforts and costs. The importance of the compressor for modern high bypass ratio engines is demonstrated by the fact that 50%–60% of the engine length, up to 40%–50% of the weight, 35%–40% of the manufacturing costs, and 30% of the maintenance costs are taken up by the compression system. The advances in compressors allow aeroengines to operate with core engine thermal efficiency in the range of 50% while the propulsive efficiency approaches 80%.

Definition - The Geometry



Axial compressors are those in which the air flows mainly parallel to the rotational axis. Axial flow compressors have a large mass flow capacity, high reliability, and high efficiency, but have a smaller pressure rise per stage (1.1:1–1.4:1) than centrifugal compressors (4:1–5:1). However, it is easy to link together several stages and produce a multistage axial compressor having pressure ratios up to 40:1.

Rotor and Stator Frames of Reference



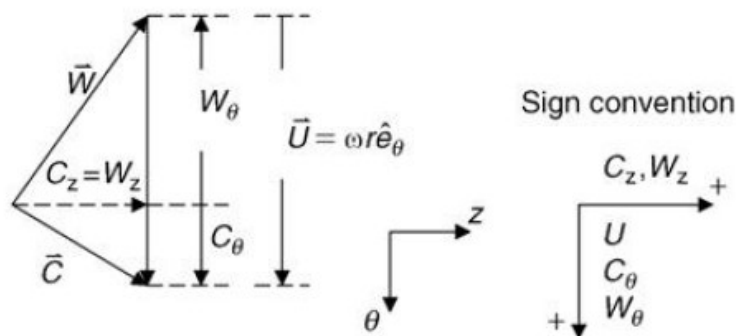
Relative swirl = Absolute swirl - ωr

Velocity Triangle

$$\vec{U} = \omega r \hat{e}_\theta$$

$$\vec{W} = \vec{C} - \vec{U}$$

$$\vec{C} = \vec{W} + \vec{U}$$



$$W_\theta = C_\theta - U$$

W_θ is the *relative* swirl
 C_θ is the *absolute* swirl

The Euler Turbine Equation

Fundamentals of Turbomachinery

The Euler's Equation

- The Euler turbine equation is called the fundamental equation of turbomachinery, this is derived as follows from the shaft power:

$$\dot{\mathcal{P}} = \omega \cdot \underbrace{\dot{m}(r_2 C_{\theta 2} - r_1 C_{\theta 1})}_{\tau_{fluid}}$$

- The ratio of shaft power to mass flow rate is called the specific work of the rotor

$$w_{c,t} \equiv \frac{\dot{\mathcal{P}}_{c,t}}{\dot{m}_{c,t}} = \omega \Delta(r C_{\theta})_{c,t}$$

- The first law of thermodynamics applied to a steady and adiabatic process demands that the change of total enthalpy of the fluid across the blade row to be equal to the blade-specific work delivered to the fluid

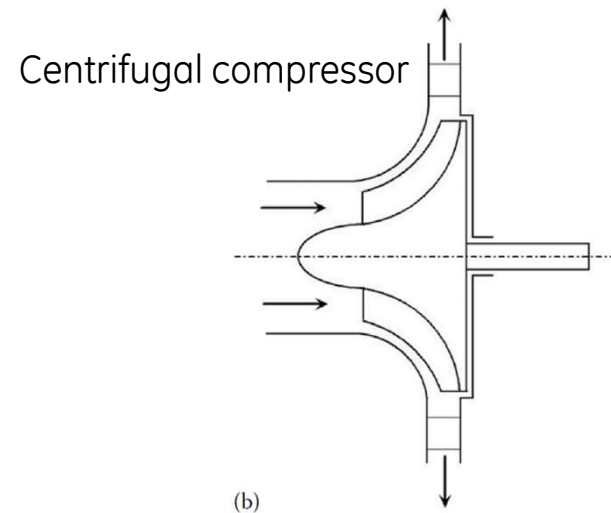
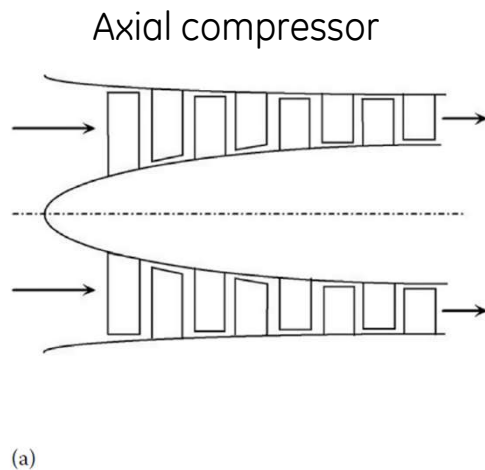
$$h_{t2} - h_{t1} = \frac{\dot{\mathcal{P}}}{\dot{m}} = w_c \quad \longrightarrow \quad \frac{h_{t2}}{h_{t1}} = 1 + \frac{\omega \Delta(r C_{\theta})}{h_{t1}} = \frac{T_{t2}}{T_{t1}}$$

Axial – Flow vs Centrifugal/Radial – Flow

Comparison

Comparison

- The centrifugal compressor achieves part of the compression process by causing the fluid to move outward in the centrifugal force field produced by the rotation of the impeller. Thus, the pressure rises due to the change in potential energy of the fluid in the centrifugal force field of the rotor. Conversely, the pressure rise in the axial flow compressor rotors and stators is achieved by the exchange of kinetic energy with thermal energy in a diffusion process.



Comparison

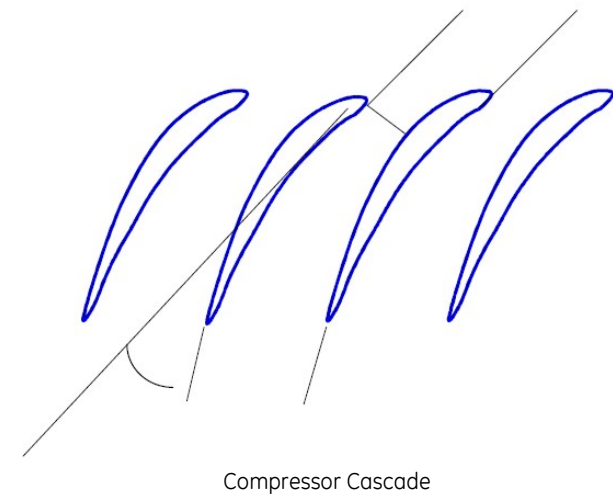
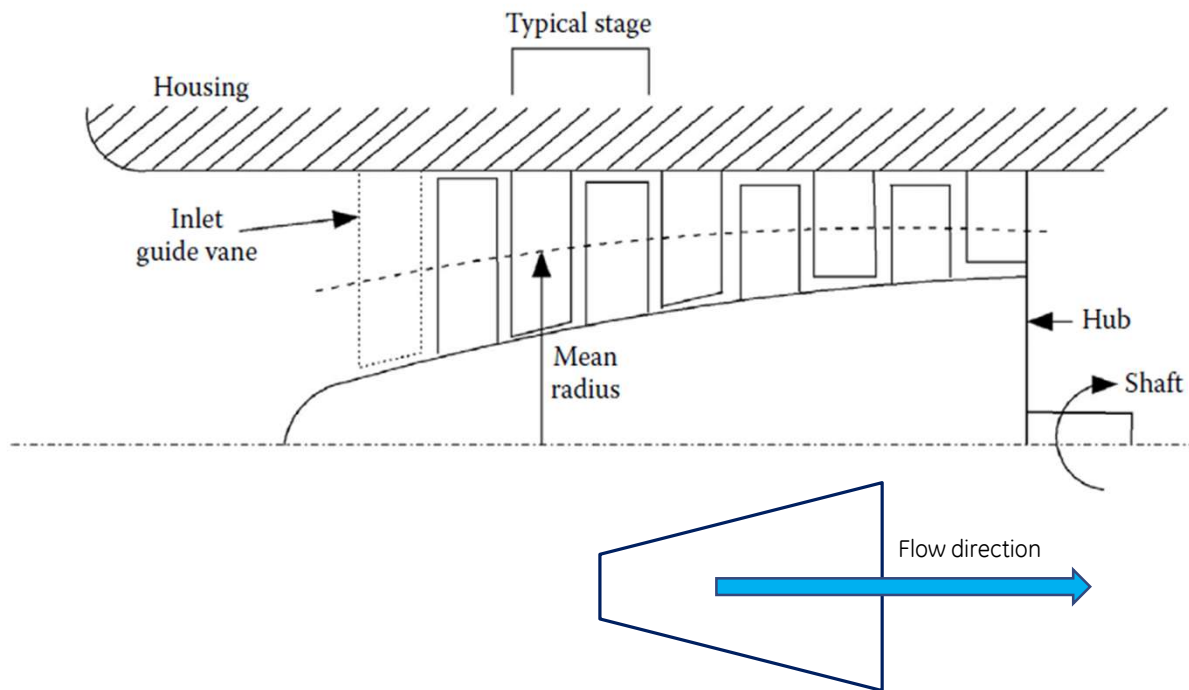
- Pros for the axial flow compressor
 - Smaller frontal area for a given mass flow rate, therefore less aerodynamic drag and nacelle housing.
 - Increased mass flow rates (particularly for high bypass ratio turbofans).
 - Flow direction discharge is more suitable for viscous effects mitigation.
 - Higher maximum pressure ratios, and higher achievable efficiencies at similar pressure ratios.
- Pros for the centrifugal flow compressor
 - Higher stage pressure ratio
 - Simplicity and less severe stall characteristics
 - Smaller dimensions for similar pressure ratios
 - Better manufacturability

Mean Flow

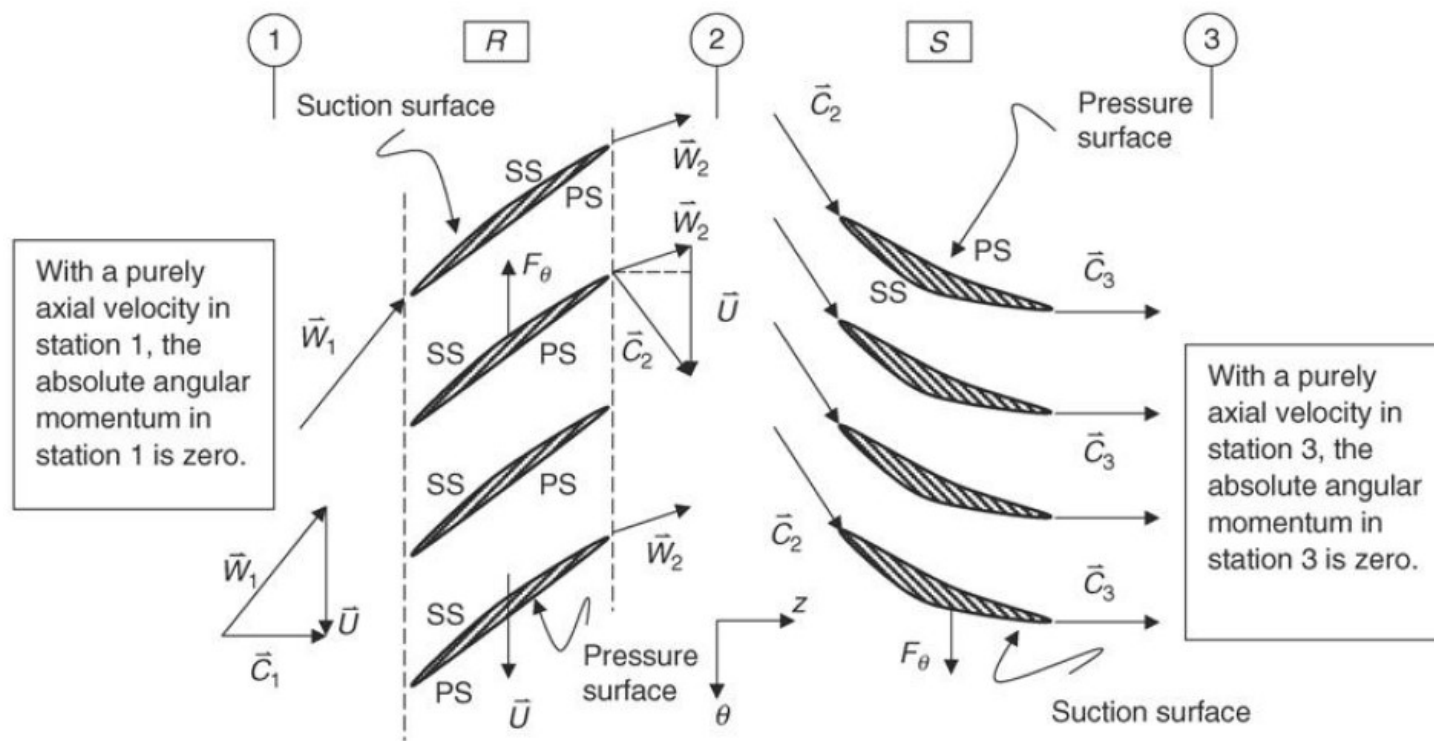
Two-Dimensional Approach

The Diffusion Process

- In very simple terms, the diffusion process is used to 'slow the flow down'. This in turn requires the flow sub-sonically to experience an increase in cross sectional area, the consequent deceleration giving rise to an increase of static pressure



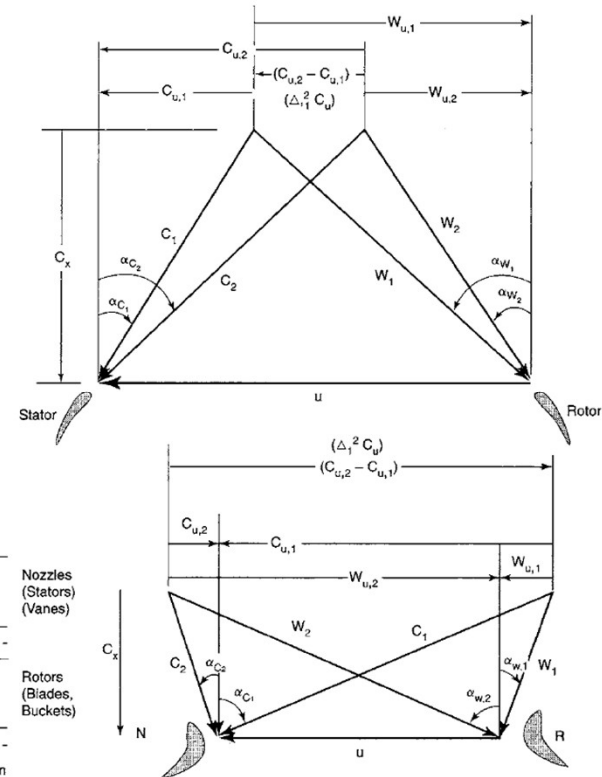
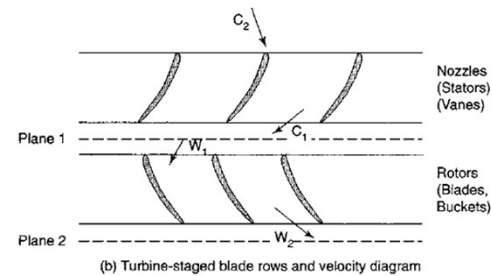
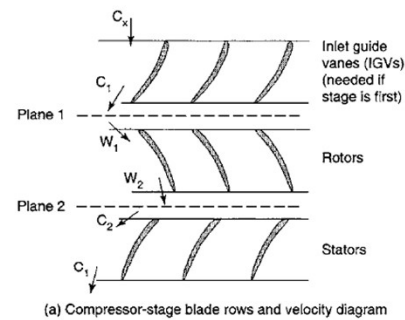
Mean flow physics



Mean Flow physics

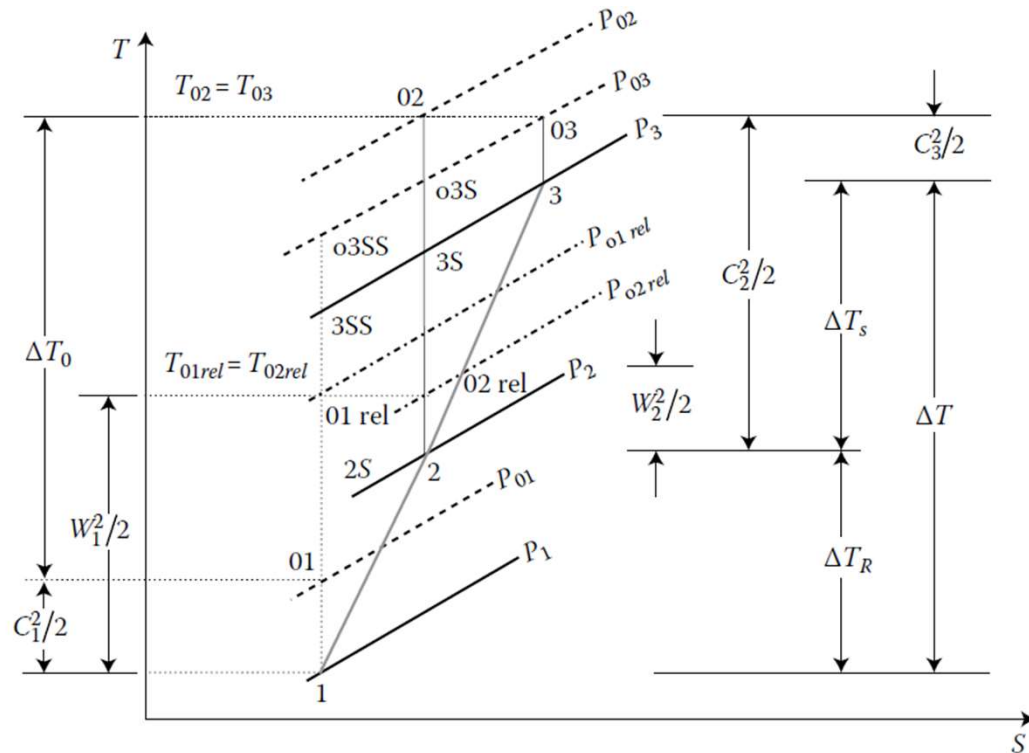
• Velocity Diagrams

- Representative axial – compressor and axial – turbine stage blade rows and simple (constant u and C_x) velocity diagrams



Mean Flow Physics

- Mollier diagram
 - the temperature–entropy variation for the compression process through the stage. If the whole processes were isentropic, the final stagnation pressure for the same work input would be P_{0max} . However, due to losses in the rotor and stator, the pressure at the outlet of the rotor is less than that maximum value and so the pressure for air leaving the stator is also less than the pressure at the outlet of the rotor, or $P_{03} < P_{02} < P_{0max}$.

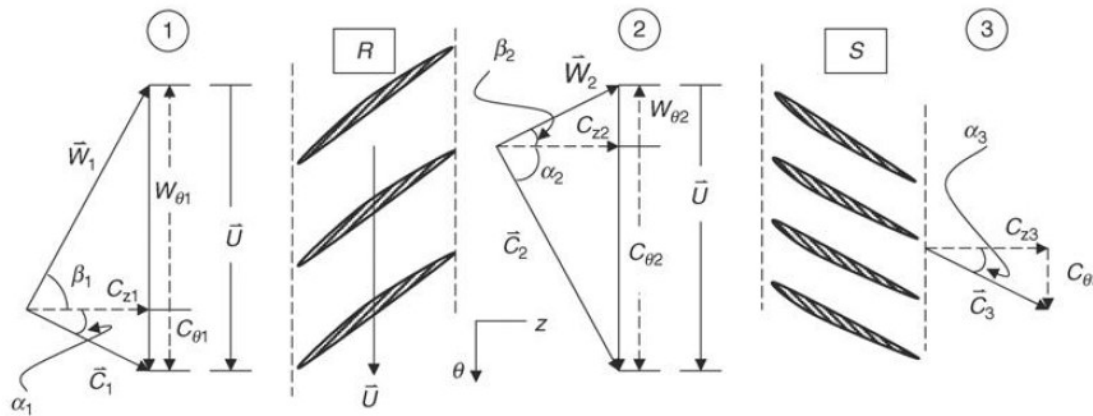


Example 8.1

- The absolute flow at the pitch line to a compressor rotor is swirl free. The exit flow from the rotor has a positive swirl, $C_{\theta 2} = 145 \text{ m/s}$. The pitch line radius is $r_m = 0.5 \text{ m}$, and the rotor angular speed is $\omega = 5600 \text{ rpm}$. Calculate the specific work at the pitch line and the rotor torque per unit mass flow rate.

Definition of Flow Angles

- The flow angles are measured with respect to the axial direction, or axis of the machine, and are labeled as α and β , which correspond to the absolute and relative flow velocity vectors $\vec{C}$ and $\vec{W}$, respectively.



$$\begin{aligned} C_{\theta1} &= C_{z1} \cdot \tan \alpha_1 \\ W_{\theta1} &= C_{z1} \cdot \tan \beta_1 \\ C_{\theta2} &= C_{z2} \cdot \tan \alpha_2 \\ W_{\theta2} &= C_{z2} \cdot \tan \beta_2 \end{aligned}$$

Definition of Flow Angles

- Repeated Stage
 - Implies that the velocity vectors at the exit and the entrance to a stage are the same

$$\begin{aligned}\vec{C}_3 &= \vec{C}_1 \\ \alpha_3 &= \alpha_1\end{aligned}$$

- Repeated row design
 - The exit flow angle has the same magnitude as the absolute inlet flow angle, and the inlet relative flow angle has the same magnitude as the exit absolute flow angle

$$\begin{aligned}|\beta_2| &= |\alpha_1| \\ |\alpha_2| &= |\beta_1|\end{aligned}$$

Stage Parameters

- Pressure ratio per stage

$$\pi_{stage} = \left[1 + \frac{\eta_{isen}}{C_p T_{t1}} U C_z (\tan \alpha_2 - \tan \alpha_1) \right]^{\frac{\gamma}{\gamma-1}}$$

- An actual stage temperature rise will be less than the above value due to the three-dimensional effects in the compressor annulus. Analysis of experimental results shows that it is necessary to multiply the results by a factor less than 1, identified as the work done factor (λ), due to the radial variation of the axial velocity

$$\pi_{stage} = \left[1 + \lambda \frac{\eta_{isen}}{C_p T_{t1}} U C_z (\tan \alpha_2 - \tan \alpha_1) \right]^{\frac{\gamma}{\gamma-1}}$$

Stage Parameters

- Flow coefficient

$$\phi \equiv \frac{C_z}{U}$$

Physically the flow coefficient parameter shows the effect of the mass flow as well as the blade speed in a compressor and/or a turbine.

In a simple velocity diagram the flow coefficient is constant and refers to the stage as a whole. In the general case, where both C_z and U vary through the stage for any stream surface, there will be a different flow coefficient at the rotor inlet and at rotor outlet, and the flow coefficient varies with radius.

What would mean a very large/very small flow coefficient?

Stage Parameters

- Stage Loading Factor/Coefficient

$$\psi \equiv \frac{\Delta h_t}{U^2}$$

The loading factor reflects pressure/temperature rise across a compressor or drop across a turbine.

In a simple velocity diagram the stage loading is a non dimensional measure of: the distance between of apexes (vertices) of the absolute and relative velocity triangles divided by U. In compressor stages highly and lowly loaded stages would have values below -0.5 and above 0.3, respectively.

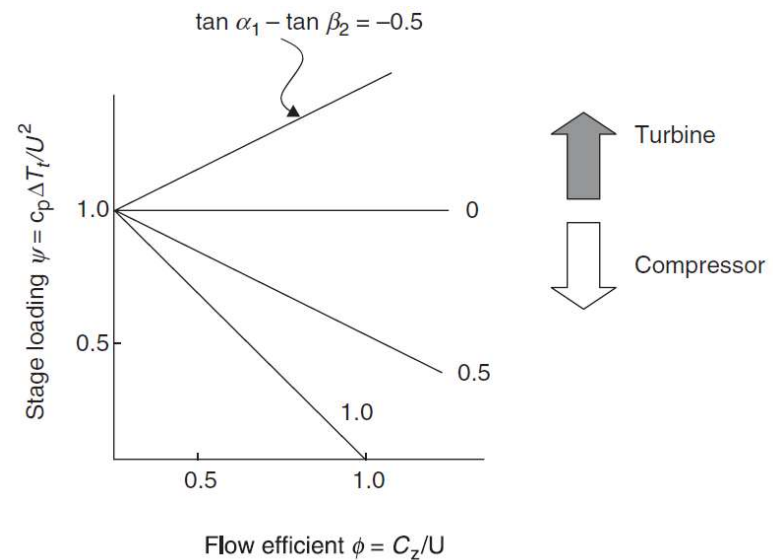
Stage Parameters

- Stage Loading and Flow Coefficient are both related via the temperature rise as follows:

$$h_{t2} - h_{t1} = \frac{\dot{Q}}{\dot{m}} = U(C_{\theta 2} - C_{\theta 1})$$

$$\psi \equiv \frac{\Delta h_t}{U^2} = \frac{C_{\theta 2}}{U} - \frac{C_{\theta 1}}{U} = \frac{W_{\theta 2} + U}{U} - \frac{C_{\theta 1}}{U} = 1 + \left(\frac{C_z}{U}\right)(\tan \beta_2 - \tan \alpha_1)$$

$$\psi = 1 + \phi(\tan \beta_2 - \tan \alpha_1)$$

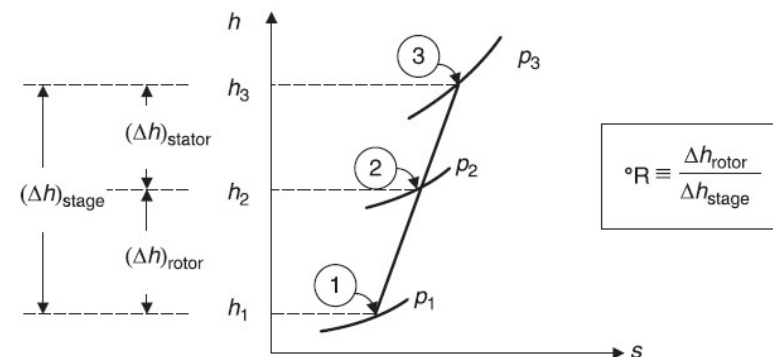


Stage Parameters

- Stage Reaction

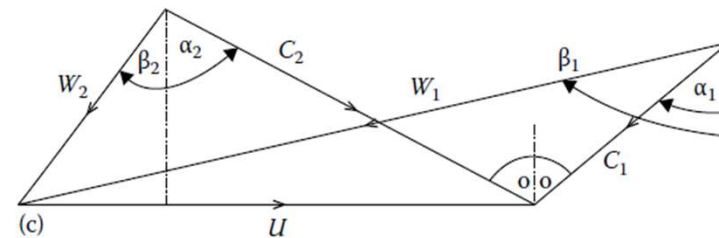
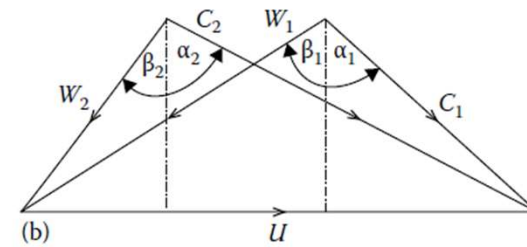
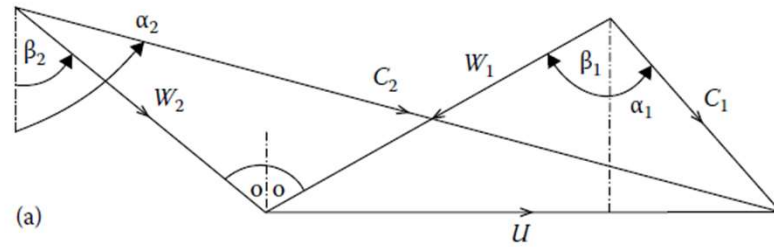
$$^{\circ}R \equiv \frac{h_2 - h_1}{h_3 - h_1} = \frac{h_{t2} - h_{t1} - (C_2^2 - C_1^2)/2}{h_{t2} - h_{t1} - (C_3^2 - C_1^2)/2}$$

- The strict definition of reaction is the ratio of change in static enthalpy to the change in stagnation enthalpy of the flow passing through the rotor.
- Although the stator does no work on the fluid, it still acts as a diffuser that decelerates the fluid and thus causes an increase in fluid temperature, or static enthalpy. The stator takes out the swirl (kinetic energy) put in by the rotor and thus converts it to a static pressure rise. The degree of reaction measures the rotor share of the stage enthalpy rise as compared with the burden on the stator.

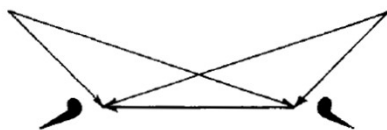


Stage Parameters

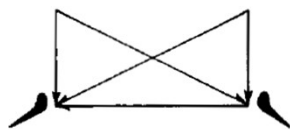
- Velocity triangles for different degrees of reaction.
 - 0%
 - 50% $\alpha_1 = \beta_2, \alpha_2 = \beta_1$
 - 100% $\alpha_1 = \alpha_2,$



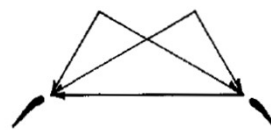
Stage Parameters



(a) High-work turbine
($\psi = 2.0, \phi = 0.5$)



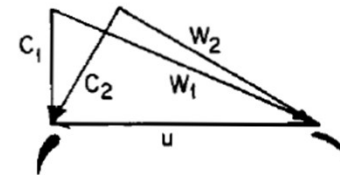
(b) Medium-work turbine
($\psi = 1.0, \phi = 0.5$)



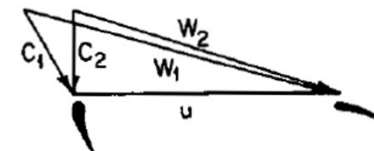
(c) High-work compressor
($\psi = -0.50, \phi = 0.375$)

Symmetric (50% reaction) velocity diagrams

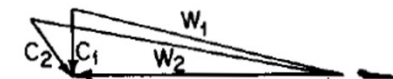
High Reaction Compressor, fan and windmill velocity diagrams



(a) Rotor-stator compressor or fan
under 100% reaction ($\psi = -0.31, \phi = 0.5$)



(b) Stator-rotor compressor or fan
over 100% reaction ($\psi = -0.25, \phi = 0.37$)



(c) Windmill rotor - no stator
over 100% reaction ($\psi = 0.1, \phi = 0.17$)

Stage Parameters

- Diffusion factor

Non dimensional parameter that relates the maximum adverse pressure gradient on the blade suction surface and the maximum flow diffusion.

$$D \equiv \frac{W_{\max} - W_2}{W_1} = \frac{W_1 + (|\Delta C_\theta|/2\sigma) - W_2}{W_1} = 1 - \frac{W_2}{W_1} + \frac{|\Delta C_\theta|}{2\sigma W_1}$$

where σ defines the blade solidity, that is, the ratio of blade chord c to spacing s

$$\sigma_r \equiv \frac{c_r}{s_r} \quad (\text{rotor solidity})$$

$$\sigma_s \equiv \frac{c_s}{s_s} \quad (\text{stator solidity})$$

Modern compressor design utilizes a high solidity ($\sigma_m \geq 1$) blading at the pitchline radius (i.e., r_m). Since the blade spacing increases linearly with radius, the solidity of a constant chord blade also decreases linearly with the blade span, that is,

$$s(r) = \frac{2\pi \cdot r}{N_b}$$

where the N_b is the number of blades in the rotor or the stator row.

Stage Parameters

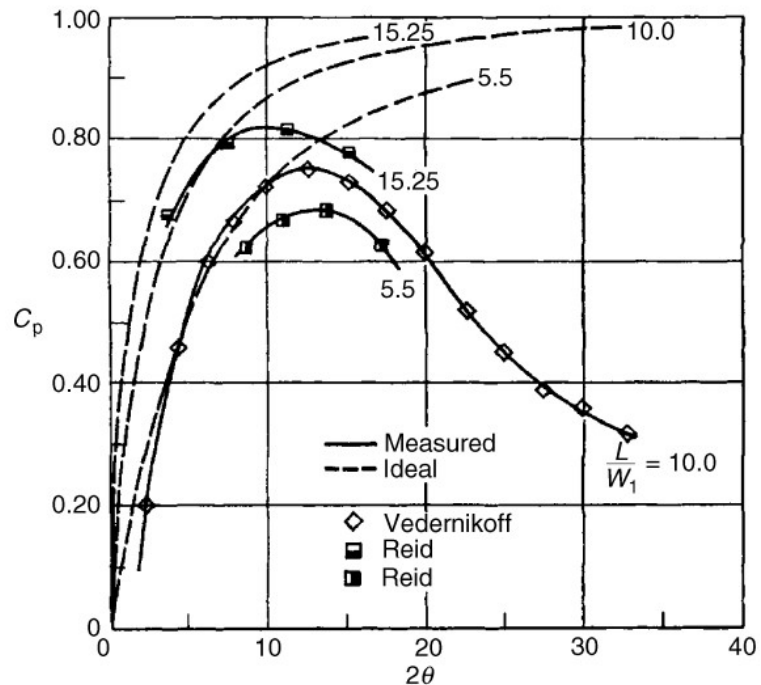
- Pressure Rise Coefficient and De Haller Number
- The pressure rise coefficient and the D -factor are two different ways of measuring the same thing, namely, the tendency of the boundary layer to stall. However, the D -factor examines the stall behavior of a compressor blade suction surface, whereas the stalling pressure rise coefficient analogy with a diffuser examines the end wall stall that dominates the multistage compressor aerodynamics.
- The new limiting parameter is defined as

$$C_p \equiv (p_2 - p_1) / (\rho_1 W_1^2 / 2)$$

- The static pressure rise across a blade row, in the numerator of previous equation, is independent of the observer frame of reference. The dynamic pressure at the inlet to the blade row is in the relative frame, that is, as in a cascade parameter.

Stage Parameters

- The main differences between a diffuser flow and a flow in a compressor blade row are
 - a) Compressor boundary layer is highly skewed due to large streamwise and radial pressure gradients, that is, leading to (a highly) three-dimensional flow separation
 - b) Inherent unsteadiness in turbomachinery due to relative motion of neighboring blade rows are absent in a diffuser
 - c) Compressibility effects through the passage shock interaction with the boundary layer is absent in diffuser performance charts
 - d) Upstream wake transport in downstream blade rows is also absent in diffusers
 - e) Blade tip clearance flow is unique to turbomachinery
 - f) End wall regions create secondary vortex formations as in corner and scraping vortex structures in compressors with no counterpart in a diffuser

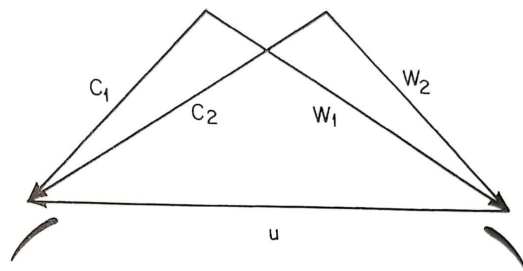


Stage Parameters influence on design

- Axial compressors are composed of series of rotating and stationary blade rows, each of which has a relative velocity at entry and a lower relative velocity at exit. Each blade row is therefore a group of parallel diffusers. (Although it is stated in the previous slides the differences).
- The principal rule on axial compressor design, is that the outlet to inlet relative velocity ratio (De Haller number) should be above some minimum value, although this should not be treated as a limit but rather as an independent, or input variable. The designer chooses a value that is estimated to give a safe margin from stall in the normal working range under the conditions the machine is likely to encounter in practice. Thus, a value of 0.72 – 0.75 might be well selected.
- The second variable that is significant in certain compressor stages is the inlet relative velocity to either stator or rotor. The first stage of a multistage air compressor is usually designed to a maximum relative Mach Number, which is found at the tip section of the rotor blades or possibly at the root or inner diameter section of the stator blades.

Stage Parameters influence on design

- The blade row velocity ratio and the maximum inlet relative velocity often become the determinants of both blade speed and the diagram shape. The remaining degree of freedom is expressed alternatively as reaction, or flow coefficient, or blade stagger angle as shown below:



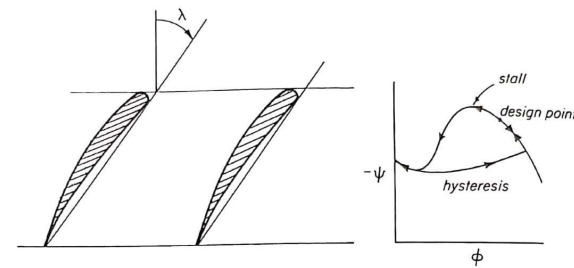
$$\phi = 0.4$$

$$\psi = -0.3$$

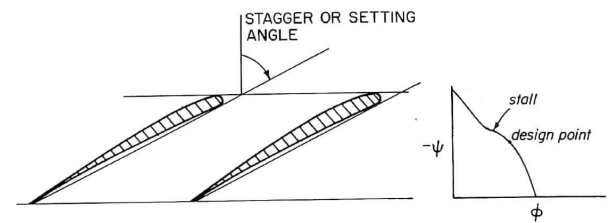
$$R_n = 0.5$$

$$\frac{W_2}{W_1} = \frac{C_1}{C_2} = 0.696$$

$$C_{pr, th} \equiv (\Delta p/q) = 0.515$$



(a) Low stagger, high flow coefficient

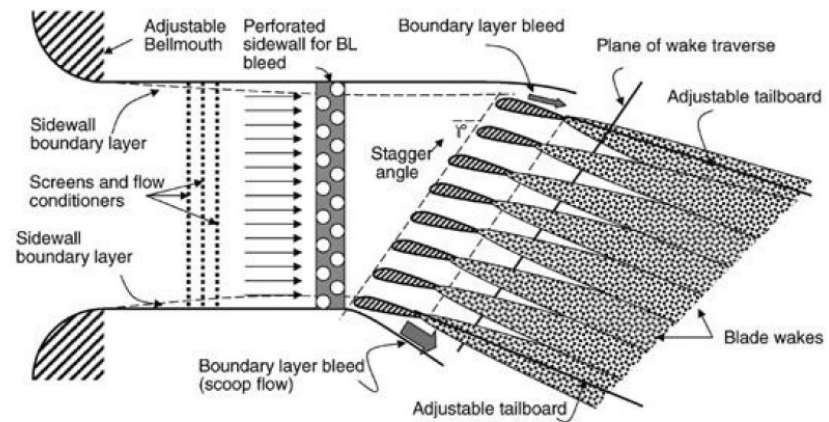


(b) High stagger, low flow coefficient

Compressor Cascade

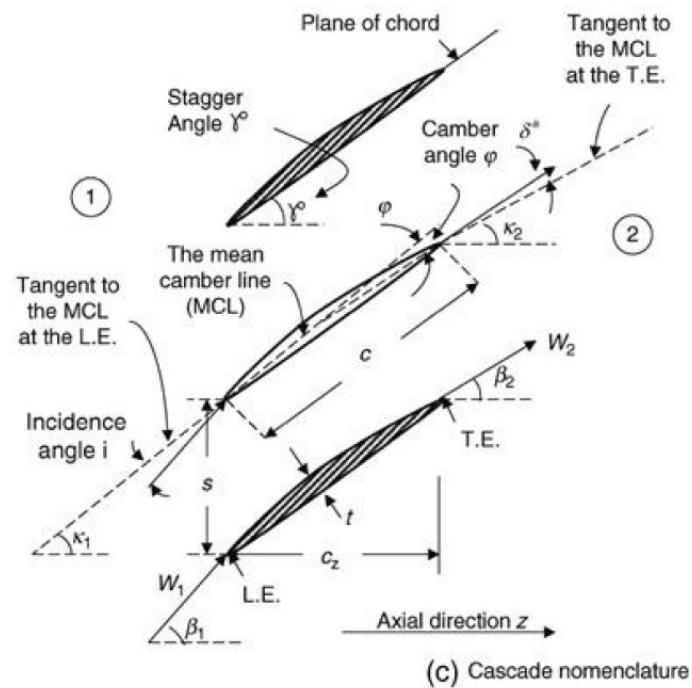
What is a cascade?

- A cascade is composed of a series of two-dimensional or cylindrical blades placed in a uniform flow to simulate the two-dimensional flow field about a spanwise section of a compressor blade row.
- The goal of the investigation is to quantify the profile loss of two dimensional blade elements as a function of the profile geometry and cascade parameters such as blade chord-to-spacing ratio, blade stagger angle, and other parameters. To construct the three-dimensional loss characteristics of compressor blades, two-dimensional cascade loss data are stacked up to represent the 3D picture.



(a) Schematic drawing of a rectilinear cascade test rig with cylindrical blades

Compressor Cascade



Cascade parameters

Solidity
 $\sigma = c/s$

Stagger angle
 γ°

Camber angle
 $\phi = \kappa_1 - \kappa_2$

Incidence angle
 $i = \beta_1 - \kappa_1$

Deviation angle
 $\delta^* = \beta_2 - \kappa_2$

Net turning angle
 $\Delta\beta = \beta_2 - \beta_1 = \phi + i - \delta^*$

Angle of attack
 $\beta_1 - \gamma^\circ$

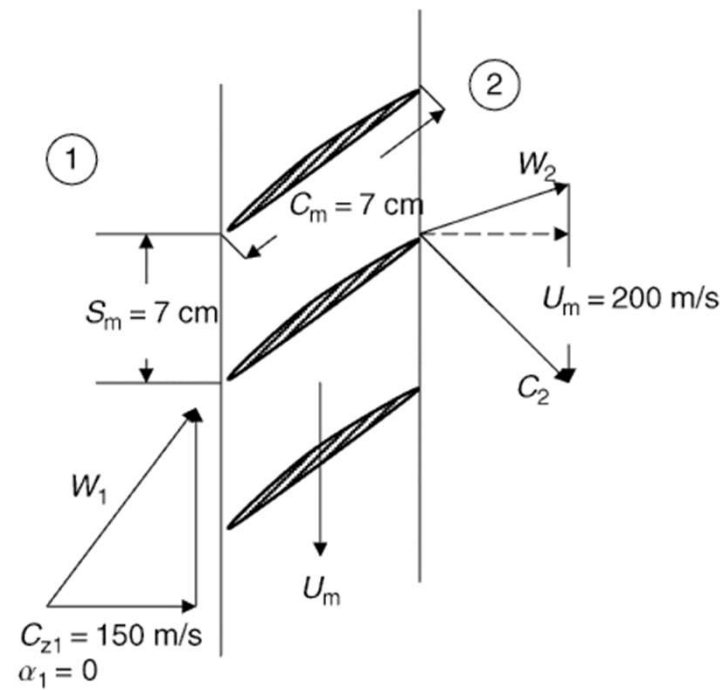
Thickness-to-chord ratio
 t/c

Example 8.2

- An axial-flow compressor stage has a pitchline radius of $r_m = 0.5$ m. The rotational speed of the rotor at pitchline is $U_m = 212$ m/s. The absolute inlet flow to the rotor is described by $C_{zm} = 155$ m/s and $C_{\theta 1m} = 28$ m/s. The stage degree of reaction at pitchline is $^{\circ}R_m = 0.60$, $\alpha_3 = \alpha_1$, and C_{zm} remains constant. Calculate:
 - a) rotor angular speed ω in rpm
 - b) rotor exit swirl $C_{\theta 2m}$
 - c) rotor specific work at pitchline w_{cm}
 - d) relative velocity vector at the rotor exit
 - e) rotor and stator torque per unit mass flow rate
 - f) stage loading parameter at pitchline ψ_m
 - g) flow coefficient ϕ_m .

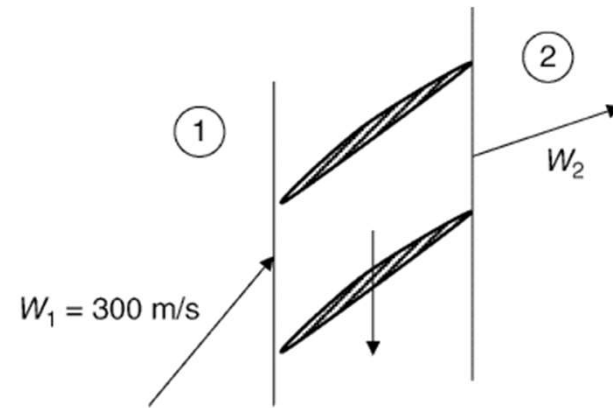
Example 8.3

- A rotor blade row is cut at pitchline, r_m . The velocity vectors at the inlet and exit of the rotor are shown.
 - Assuming that $U_{m1} = U_{m2} = 200 \text{ m/s}$ and $C_{z1} = C_{z2} = 150 \text{ m/s}$ and $\beta_2 = -35^\circ$, calculate
 - W_{1m} and W_{2m}
 - D-factor D_{rm}



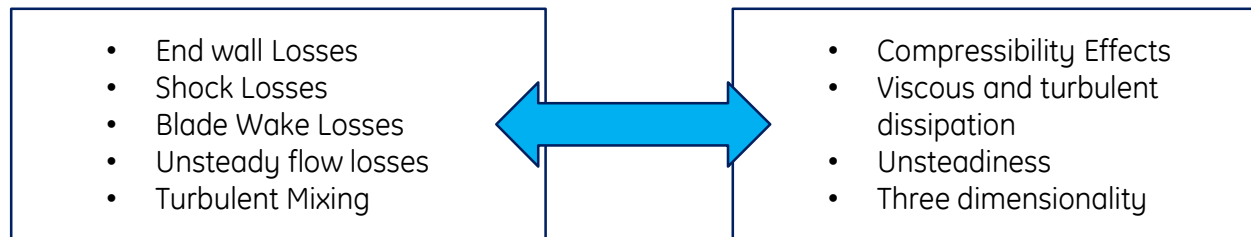
Example 8.4

- A rotor blade row at pitch line radius is shown. The rotor total pressure loss coefficient at this radius is $\varpi_{rm} = 0.03$.
 - a) How much deceleration is allowed in the rotor under de Haller criterion? i.e., What is the minimum W_2 ?
 - b) What is the static pressure rise coefficient, assuming incompressible flow and W_{2min} from de Haller criterion?



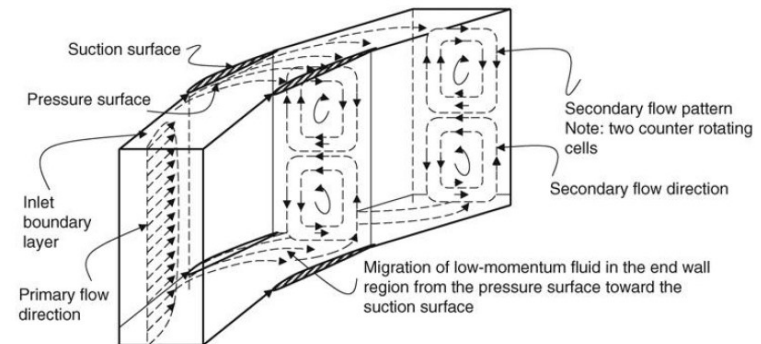
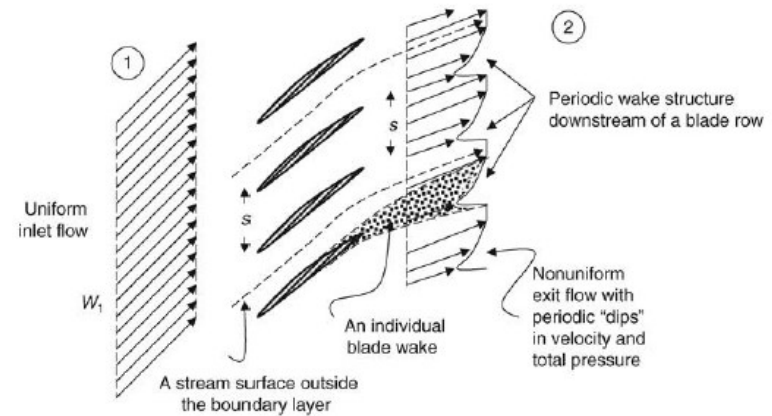
Entropy Sources: Three dimensional Losses

- The factors that render the flow process in turbomachinery stage irreversible are:



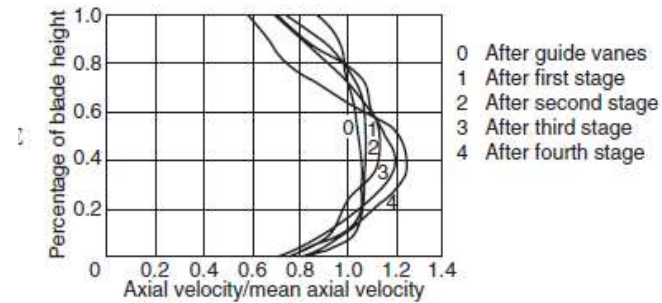
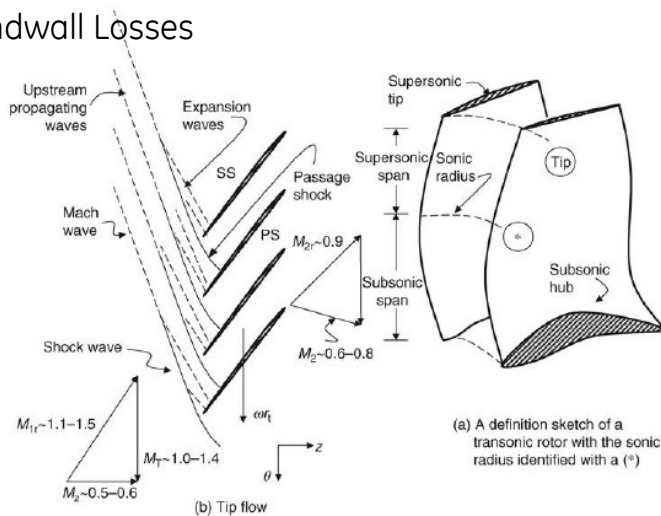
Three dimensional Losses

- Performance Prediction of axial flow compressor is manageable when we treat losses as if they were separable, when in fact they are to some extent interactive and connected.
- Group 1 Losses
 - Are pressure losses resulting from fluid friction. Examples are boundary layer friction on blade and casing walls and flow separation losses in areas of stall and blade trailing edges.
- Group 2 Losses
 - Take shaft power and degrade it to an enthalpy increase in the gas discharge condition.
- Group 3 Losses
 - Losses of the shaft power through friction, the energy from which is dissipated away from the working fluid. Examples are the friction losses in external bearing and seals, thus we can designate them as "external losses".

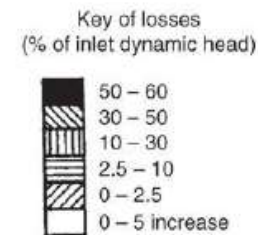
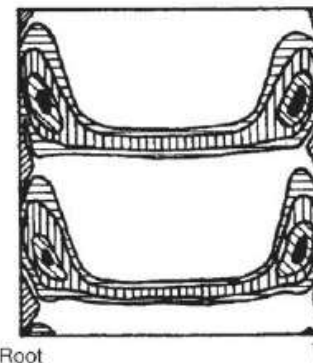
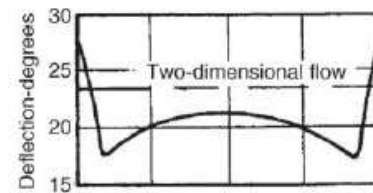


Group 1 Losses

- Blade profile losses
- High Mach Number leading edge losses
- Endwall Losses



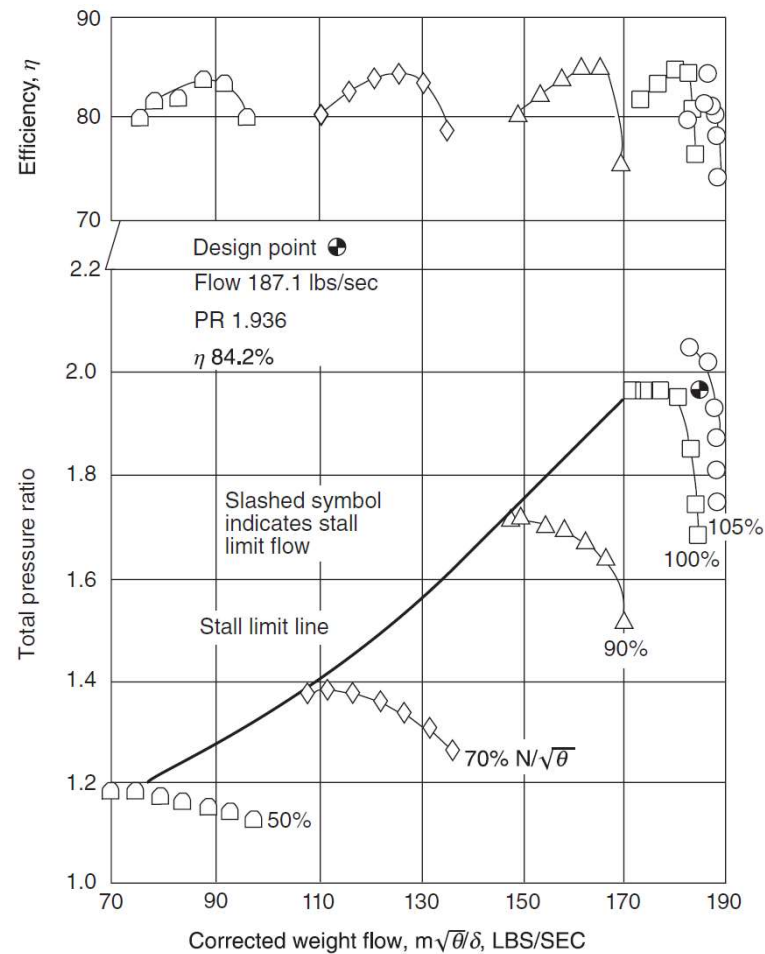
(a) The evolution of axial velocity profile with successive blade rows in a multistage compressor



(b) Total pressure loss distribution with blade height in a cascade

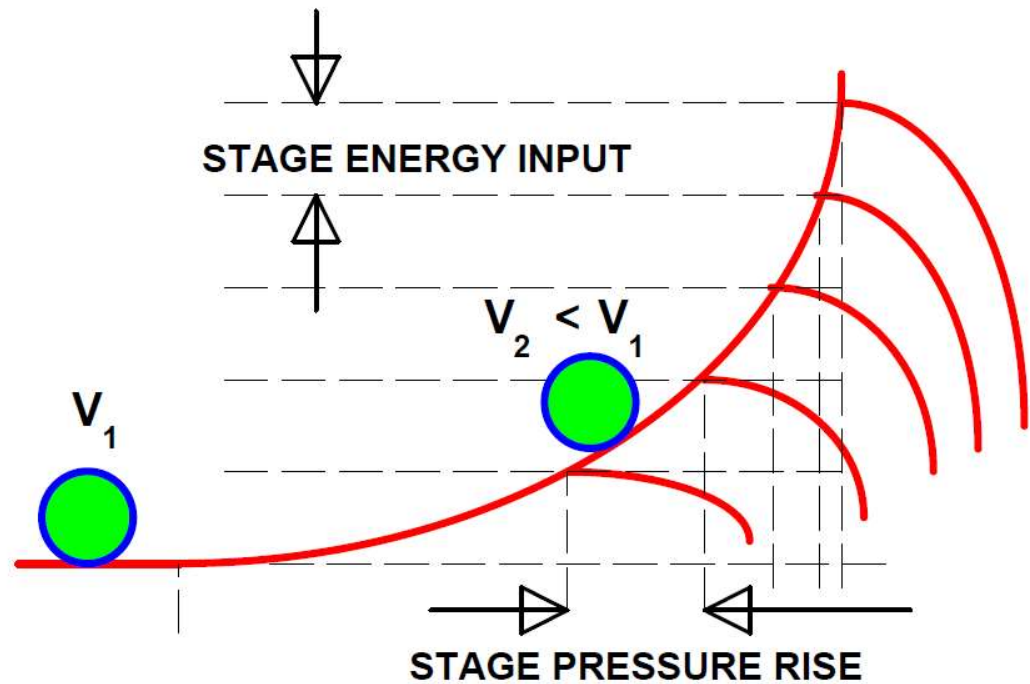
Compressor Performance Map

- Compressor pressure ratio plotted against the mass flow rate through the compressor is the compressor performance map. It is customary to graph the constant rpm lines on the performance chart as well as the adiabatic efficiency.



Compressor Performance Map

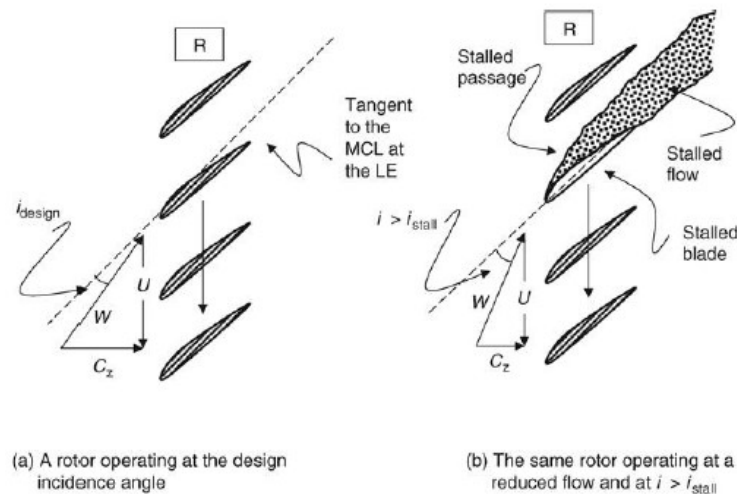
- The easiest way to visualize/interpret the compressor performance map is by understanding the meaning of the variables involved in it.



Compressor Instability – Stall and Surge

Compressor Stall

- In a compressor operating at a constant rotational (shaft) speed when the mass flow rate drops, the axial velocity decreases, hence the incidence angle increases.
- Large deviation from this minimum-loss incidence angle causes a rapid rise in total pressure loss, which signifies boundary layer separation.



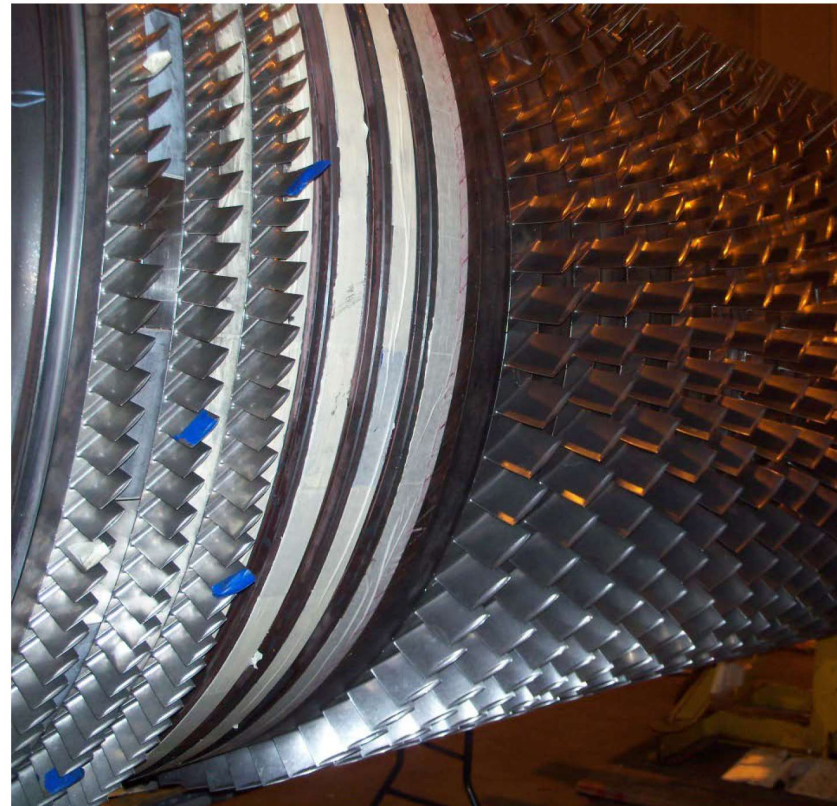
Compressor Instability – Stall and Surge

Surge

- The coupling between the compressor stall instability and the combustion chamber resonant characteristics could lead to an overall breakdown of the flow, or flow oscillation, in the compressor and combustor. The overall flow breakdown in the “system” composed of compressor and combustor is called surge. In a fully developed stage, surge is an axisymmetric oscillation of the flow with a characteristic timescale dictated by the plenum chamber time to empty and fill. In the initial transient stage, surge is asymmetric and thus creates transverse loads on the blades, which may rub on the casing and lead to structural damage and system breakdown.

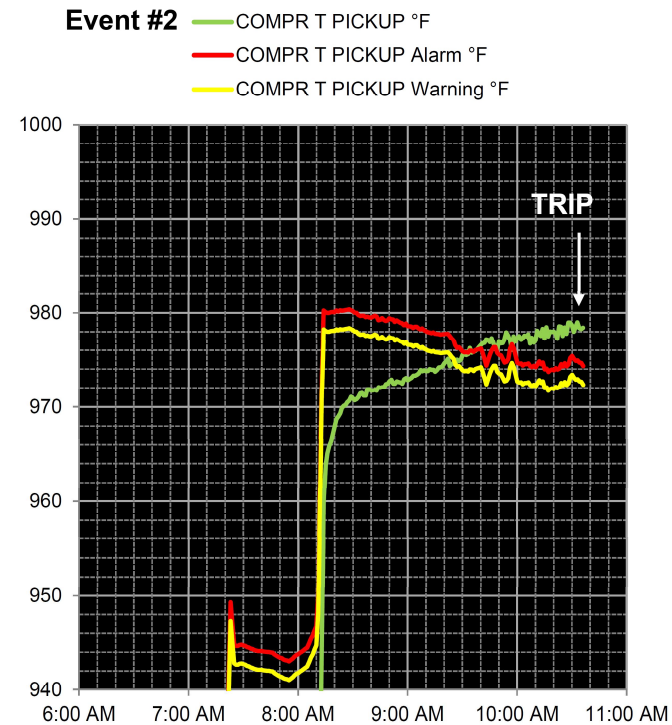
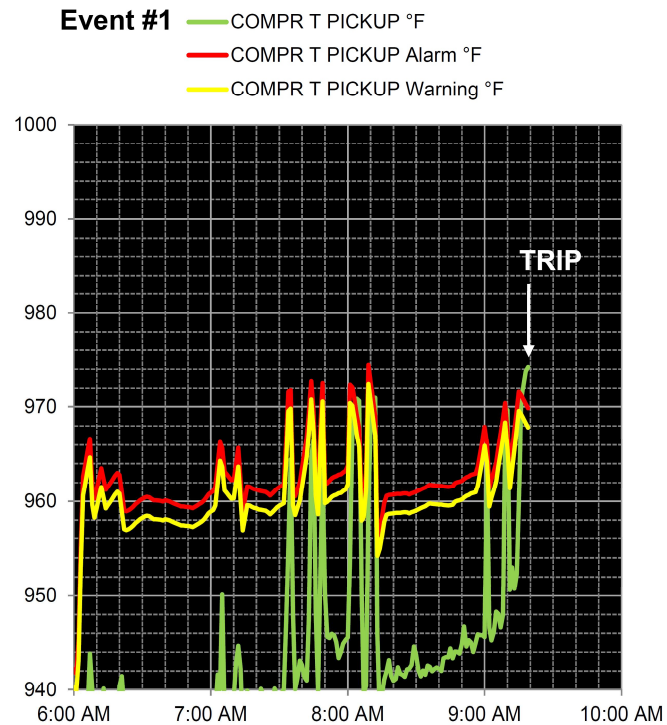
Compressor Diagnostics

- Two compressor failures (less than 12 months apart)
- Both events caused sudden trips with little/no warning from control systems alarms.
- Both events required replacement of compressor blades, especially later stages.



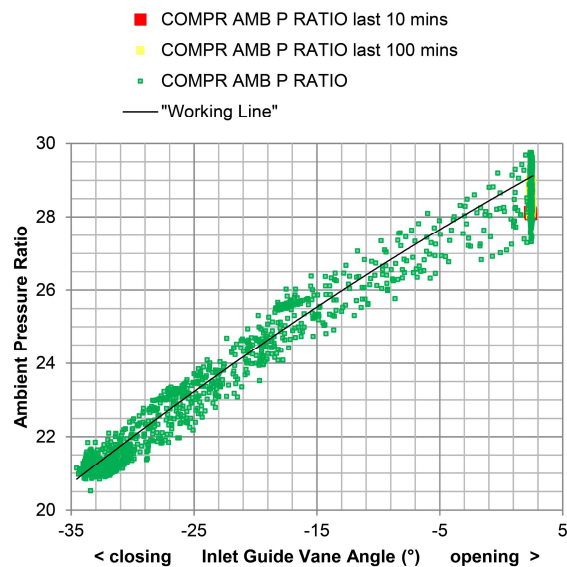
Compressor Diagnostics

- Data of such events

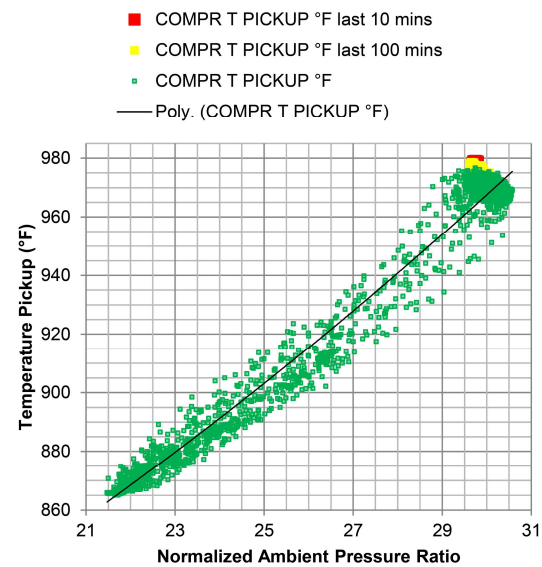


Compressor Diagnostics

- Compressor Map vs Temp Pickup



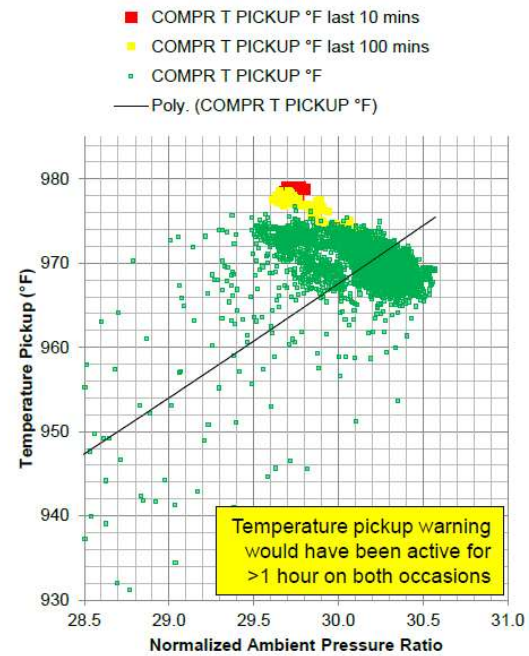
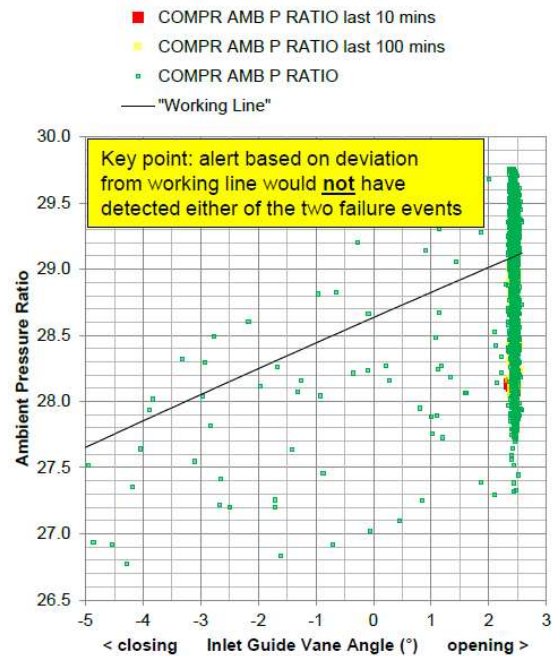
Compressor map shows no exceptional behavior before trip. Surge line would be above the working line!



Temp pickup shows exceptional behavior prior to the trip. See next slide for zoomed views.

Compressor Diagnostics

- Compressor Map vs Temp Pickup



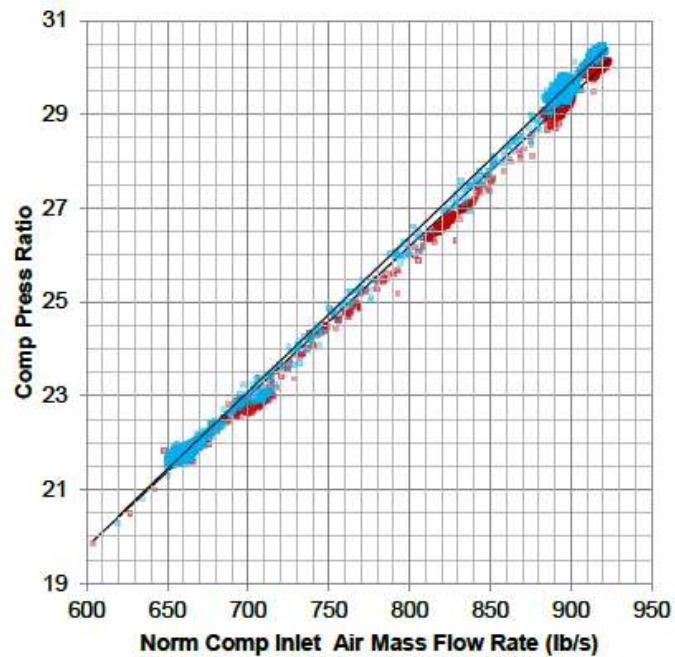
Compressor Diagnostics

- Compressor Map after Water Wash

Red = before
Blue = after

Slight change
in the trendline
(working line)

Press ratio has
increased by 0.35
(1.2%) at 920 lb/s



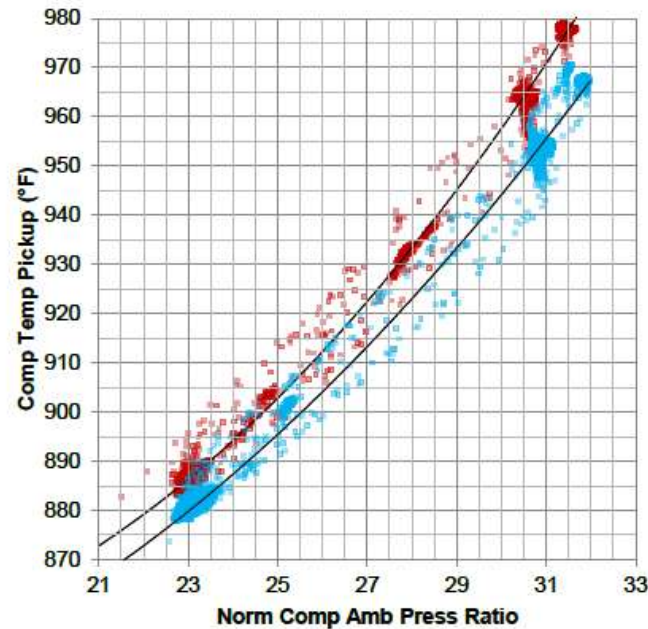
Compressor Diagnostics

- Compressor Map after Water Wash

Red = before
Blue = after

Significant change
in the trendline

At highest press
ratios, pickup has
reduced by $>15^{\circ}\text{F}$



Main effect of
offline wash
is therefore a
reduction in
aerofoil skin
friction and hence
reduction in temp
pickup

Compressor Design Parameters and Principles

Compressor Design Parameters and Principles

- The following approach to compressor design is outlined in steps that are particularly useful for students who want to learn and practice their turbomachinery design skills.
- Here, the design of the axial compressor is done at the pitchline. The process is started by choosing:
 1. Axial Mach number at the compressor/fan face is M_{z1}
 2. Flow area at the compressor face is $A_1 = \pi r_{tl}^2 - \left(\frac{r_{h1}}{r_{tl}}\right)^2$ that sizes the engine and thus establishes the mass flow rate
 3. The relative tip Mach number M_{r1} establishes the tip radius $M_{r1} = \sqrt{M_{z1}^2 + M_{t1}^2}$
 - M_{t1} , tip tangential Mach number, that is, $(\omega r_t)/a_1$ is in excess of sonic
 - M_{r1} , is supersonic, ~1.2 to ~1.5
 4. Thickness to chord ratio varies from ~ 10% in the subsonic hub region to ~3% in the supersonic tip region, linearly varying in between

Compressor Design Parameters and Principles

5. The Reynolds number based on chord has to be $> 300k$ (in relative frame), at altitude. This establishes the minimum chord length for turbulent boundary layer at all altitudes, the other parameter contributing to the chord length is the Blade aspect ratio and bending stresses. Often chord length is several times bigger than that required for the $300k$ Reynolds number at altitude.
6. Centrifugal stress is a function of centrifugal force and the blade cross sectional area given by

$$\frac{\sigma_c}{\rho_{blade}} = \frac{\omega^2 A}{4\pi} \left(1 + \frac{A_t}{A_h} \right)$$

- The blade area distribution along the span, is known as taper and is often approximated to be a linear function of the span. Therefore, it may be written as

$$\frac{A_b}{A_h} = 1 - \frac{r - r_h}{r_t - r_h} \left(1 - \frac{A_t}{A_h} \right)$$

Compressor Design Parameters and Principles

7. The solidity at the pitchline $\sigma_m \sim 1$ to 2
8. The degree of reaction at the pitchline varies depending on whether the section is subsonic or supersonic and the heat duty of the compressor. The Euler turbine equation is used to establish the rotor-specific work at the pitchline as well as the stagnation temperature rise, that is, we get T_{t2m} . The choice of σ is less critical than D-factor or de Haller number.
9. The D-factor at the pitchline should be the one with acceptable level of diffusion, using the de Haller criterion that limits such diffusion.
10. The polytropic efficiency is assumed to be constant, which in combination with T_t ratio gives p_t ratio, that is, we get total pressure at the pitchline of the station 2, downstream of the rotor.
11. To get the flow parameters downstream of the stator, we preserve T_t and calculate p_{t3} based on an assumed ω_s ; also the idea of repeated stage gives us the α_3 .

Compressor Design Parameters and Principles

12. We assume the axial velocity remains constant throughout the compressor. This is a PRELIMINARY design choice.

Design Choices

- Hub – Shroud (Tip) Diameter ratio
 - The degree of freedom there is to choose the shape of the velocity diagrams is a strong function of the hub-shroud diameter ratio, Λ , or the ratio of the inner to the outer diameter of the blade row

$$\Lambda \equiv \frac{d_{hb}}{d_{sh}}$$

- When there is a strong need to keep outside diameter as small as possible, the diameter ratio of the first blade row of an aircraft compressor or fan may be made as small as 0.3. In industrial design the optimum configuration will normally be one having a higher hub-shroud ratio at inlet perhaps of 0.7. When severe design constraints are imposed, such as a first stage hub-shroud diameter ratio of less than 0.5, “simple” velocity diagrams become, for most practical flow distributions, too approximate to be valuable.

Design Choices

- Choice of number of stages
 - The pressure ratio may be high enough to result in the requirement of a rotor peripheral speed that will lead to high blade or disk stresses. (This is a principal reason why large high pressure ratio gas turbines use multi-stage axial flow compressors and turbines).
 - The high velocities required may lead to Mach numbers that introduce high losses or reduce range to too high a degree. (In industrial gas turbines an increased number of stages in the axial flow compressor is often accepted as a worthwhile cost in return for subsonic and therefore higher-efficiency initial stages).
 - By going to multiple stages of lower aerodynamical loading, the outlet velocity will usually be reduced relative to a more highly loaded stage of otherwise identical specifications. (This increases the stagnation to static efficiency, which is usually the controlling factor in overall machine efficiency.)
 - By going to multiple stages of identical aerodynamical loading, the outlet velocity and kinetic energy are reduced proportionately: if two stages are used instead of one, the outlet kinetic energy is halved. (Again the stagnation to static efficiency is increased).

Design Choices

- Blade Design

- For blade design, we need to estimate the following angles:

Incidence angle i

Deviation angle δ^*

- The incidence angle is estimated at the location of minimum loss, from cascade data that is known as $i_{optimum}$. Typically, optimum incidence angle is $\sim -5^\circ$ to $+5^\circ$.
- The deviation angle is calculated based on the net flow turning and the coefficient “m” in Carter’s formula is taken to be 0.25. The formula states as follows:

$$\delta^* = m\varphi/\sqrt{\sigma}$$

- The first iteration will follow an approximation of:

$$\delta^* \approx \frac{\Delta\beta}{4\sqrt{\sigma}}$$

- Which will be followed by

$$\begin{array}{l} i \equiv \beta_1 - \kappa_1 \longrightarrow \kappa_1 \equiv \beta_1 - i \\ \delta^* \equiv \beta_2 - \kappa_2 \longrightarrow \kappa_2 \equiv \beta_2 - \delta^* \longrightarrow \varphi \equiv \kappa_1 - \kappa_2 \end{array}$$

Design Choices

- Blade Design
 - Which allows to use the initial Carter's Rule.

$$\delta^* = m\varphi/\sqrt{\sigma}$$

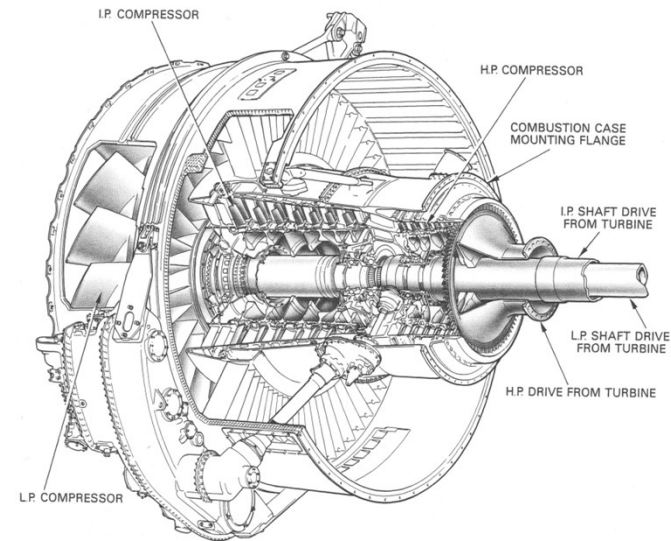
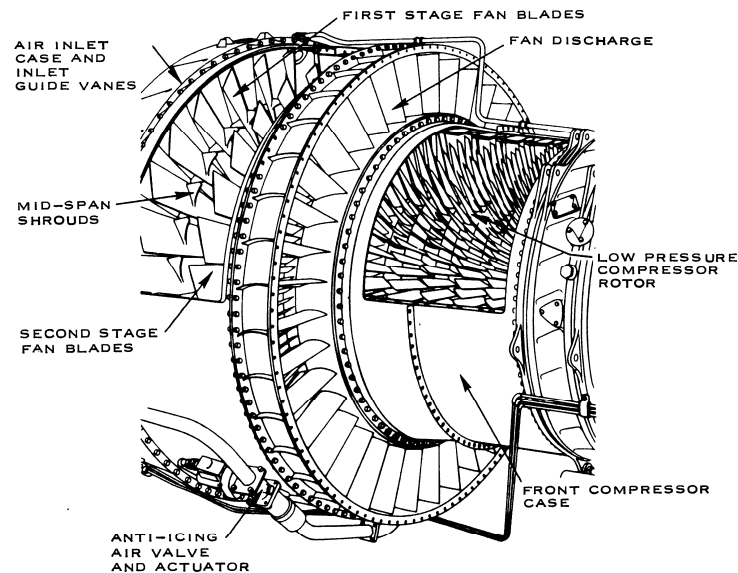
- And so on until the iteration loop converges.

Equal enthalpy change per stage is usual but not universal

- The simplest method of dividing up the enthalpy rise or fall is to have equal enthalpy changes per stage. However, there are many exceptions.
- The first stage in a transonic fan or compressor may be given a large enthalpy rise so that the second and later stages will be subsonic.

Guidelines on the Range of Compressor Parameters		
Parameter	Range of values	Typical value
Flow coefficient ϕ	$0.3 \leq \phi \leq 0.9$	0.6
D -Factor	$D \leq 0.6$	0.45
Axial Mach number M_z	$0.3 \leq M_z \leq 0.6$	0.55
Tip Tangential Mach Number, M_T	1.0–1.5	1.3
Degree of reaction	$0.1 \leq {}^\circ R \leq 0.90$	0.5 (for $M < 1$)
Reynolds number based on chord	$300,000 \leq Re_c$	>500,000
Tip relative Mach number (1st Rotor)	$(M_{1r})_{tip} \leq 1.7$	1.3–1.5
Stage average solidity	$1.0 \leq \sigma \leq 2.0$	1.4
Stage average aspect ratio	$1.0 \leq AR \leq 4.0$	<2.0
Polytropic efficiency	$0.85 \leq e_c \leq 0.92$	0.90
Hub rotational speed	$\omega r_h \leq 380 \text{ m/s}$	300 m/s
Tip rotational speed	$\omega r_t \sim 450\text{--}550 \text{ m/s}$	500 m/s
Loading coefficient	$0.2 \leq \psi \leq 0.5$	0.35
DCA blade (range)	$0.8 \leq M \leq 1.2$	Same
NACA-65 series (range)	$M \leq 0.8$	Same
De Haller criterion	$W_2/W_1 \geq 0.72$	0.75
Blade leading-edge radius	$r_{L.E.} \sim 5\text{--}10\% \text{ of } t_{max}$	5% t_{max}
Compressor pressure ratio per spool	$\pi_c < 20$	up to 20
Axial gap between blade rows	$0.23 c_z \text{ to } 0.25 c_z$	$0.25 c_z$
Aspect ratio, fan	$\sim 2\text{--}5$	<1.5
Aspect ratio, compressor	$\sim 1\text{--}4$	~ 2
Taper ratio	$\sim 0.8\text{--}1.0$	0.8

Isometric and Cutaway View of Axial Flow Compressor Hardware



Summary

- Axial-flow compressors are highly evolved machines that provide efficient mechanical compression (of the gas) in a gas turbine engine. The mechanical work is delivered to the medium by a set of rotating blade rows. The rotor blades impart swirl to the flow, thereby, increasing the total pressure of the fluid. In between the rotor blade rows are the stator blade rows that remove the swirl from the fluid, thereby increasing the static pressure of the fluid.
- The combination of one rotor row and one stator blade row is called a compressor stage. Since the primary flow direction is along the axis of the machine, staging the axial-flow compressor is easy. The flow in an axial-flow compressor is exposed to an *adverse pressure gradient* environment, that is, a climbing pressure hill that tends to stall the boundary layer. Therefore, the phenomenon of *stall* is experienced in a compressor. The stalled compressor flow is inherently unsteady and thus may cause a system-wide instability (between the compressor and combustor) known as *surge*.